Oblig 2

$f(x) = e^{-x/4} \cdot \tan^{-1} x$

$f'(x) = u'(x)v(x) + u(x)v'(x)$

Substitute.

$f'(x) = \frac{1}{4} e^{-x/4} \tan^{-1}(x) + e^{-x/4} \frac{1}{x^2+1}$

factor out $e^{-x/4}$

$f'(x) = e^{-x/4}\left(\frac{1}{x^2+1} - \frac{1}{4}\tan^{-1}(x)\right)$

~~step~~ Set derevative to zero

$f'(x) = 0$

Since $e^{-x/4} \neq 0$

$\frac{1}{x^2+1} - \frac{1}{4}\tan^{-1}(x) = 0$

multiply by 4

$\tan^{-1}(x) - \frac{4}{x^2+1} = 0$

when divided by 4

$\frac{1}{4}\tan^{-1}(x) - \frac{1}{x^2+1} = 0$

Since we need to solve the equation numerically

$g(x) = \frac{1}{x^2+1} - \frac{1}{4}\tan^{-1}(x)$

if $(x)$ is $= 0$

$g(0) = \frac{1}{0+1} - \frac{1}{4}\tan^{-1}(0)$

$g(0) = 1 - \frac{1}{4}(0)$

$g(0) = 1$ therfore $1 > 0$ so $g(0) > 0$

the function increases funksjonen øker.

$u(x) = e^{-x/4}$

$u'(x) = -\frac{1}{4} e^{-x/4}$

$v(x) = \tan^{-1}(x)$

$v'(x) = \frac{1}{1-x^2}$

When $x = 4$

$$g(x) = \frac{1}{x^2+1} - \frac{1}{4}\tan^{-1}(x)$$

$$g(4) = \frac{1}{4^2+1} - \frac{1}{4}\underbrace{\tan^{-1}(\overset{4}{\overset{\shortparallel}{x}})}_{\text{radian of } \tan^{-1}(4) \text{ is } = 1.3258}$$

$$= \frac{1}{17} - \frac{1}{4}$$

$$= \frac{1}{17} - \frac{1}{4} \times 1.3258$$

$$= -0.2726$$

a peak point.